

Chapter 5.4 – Validation of Regression Methods

Detailed classroom presentation based on the Chapter 5 material.

Validation metrics: Standard Error / Residual Standard Error, MAE, MSE, RMSE, Relative MSE, Coefficient of Variation, r^2 and Standard Error Estimate.

Includes the textbook worked validation example and additional step-by-step problems.

Textbook section: pp. 138–141.

Learning Objectives

Explain why a fitted regression model must be validated.

Calculate MAE, MSE and RMSE from actual and predicted values.

Understand Relative MSE and its interpretation.

Calculate and interpret the coefficient of variation.

Calculate coefficient of determination r^2 .

Understand explained and unexplained variation.

Calculate and interpret a standard error estimate.

Compare two regression models using validation metrics.

Select appropriate metrics for a regression application.

Why Validate a Regression Model?

A regression equation can produce predictions, but the equation alone does not tell us how accurate those predictions are.

Validation compares observed/actual response values with predicted values.

A good validation process quantifies prediction error and goodness of fit.

Different metrics emphasize different aspects of model performance.

The chapter emphasizes MAE, MSE, RMSE, ReIMSE, CV, r^2 and standard error estimate.

5.4 – Validation Measures at a Glance

Measure	What it tells us	Preferred direction
MAE	Average absolute prediction error	Lower is better
MSE	Average squared prediction error	Lower is better
RMSE	Error on original response scale	Lower is better
Relative MSE	Error relative to a reference/variability baseline	Lower is better; 0 is perfect
CV	Variability relative to the mean	Context dependent
r^2	Proportion of response variation explained	Higher is generally better
Standard Error Estimate	Residual spread around fitted relationship	Lower generally indicates tighter fit

Residuals – Starting Point for Validation

For each observation, residual $e_i = y_i - \hat{y}_i$.

Actual value y_i : observed response.

Predicted value $\hat{y}_i$: value generated by the regression model.

Residuals may be positive, negative or zero.

Validation measures summarize these residuals over all observations.

Large residuals indicate predictions that are far from observations.

Mean Absolute Error (MAE)

Textbook pp. 138–139

MAE is the average absolute prediction error.

Formula: $MAE = (1/n) \sum |y_i - \hat{y}_i|$.

Absolute values prevent positive and negative errors from cancelling.

MAE remains in the original units of the response variable.

Lower MAE indicates smaller average absolute prediction error.

MAE gives every absolute error equal weight.

Mean Squared Error (MSE)

Textbook pp. 138–139

MSE is the average of squared prediction errors.

Formula: $MSE = (1/n) \sum (y_i - \hat{y}_i)^2$.

Squaring prevents positive and negative errors from cancelling.

Large errors receive disproportionately greater weight.

MSE is expressed in squared units of the response variable.

Lower MSE indicates better fit under this metric.

Root Mean Squared Error (RMSE)

Textbook pp. 139–140

RMSE is the square root of MSE.

Formula: $RMSE = \sqrt{MSE}$.

It returns the metric to the same units as the response variable.

RMSE is particularly useful when we want error magnitude in the original measurement scale.

Lower RMSE generally indicates better predictive accuracy.

Example from the chapter: $MSE = 62.5 \rightarrow RMSE = \sqrt{62.5} \approx 7.91$.

Worked Problem 1 – Textbook Validation Example

Textbook pp. 139–140

Actual values: $y = [80, 75]$.

Predicted values: $\hat{y} = [75, 85]$.

Step 1: calculate the residuals.

Step 2: calculate MAE.

Step 3: calculate MSE.

Step 4: calculate RMSE.

This is the validation example presented in the Chapter 5 material.

Problem 1 – Step-by-Step Calculation

Observation	Actual y	Predicted $\hat{y}$	Residual e	$ e $	e^2
1	80	75	+5	5	25
2	75	85	-10	10	100
Total	155	160	-5	15	125

Problem 1 – MAE, MSE and RMSE Solution

$$\text{MAE} = (5 + 10)/2 = 15/2 = 7.5.$$

$$\text{MSE} = (25 + 100)/2 = 125/2 = 62.5.$$

$$\text{RMSE} = \sqrt{62.5} \approx 7.91.$$

Interpretation: the average absolute prediction error is 7.5 response units.

RMSE indicates a typical error scale of approximately 7.91 response units.

These are the values highlighted in the textbook validation example.

Relative MSE (RelMSE)

Textbook p. 140

Relative MSE compares prediction error with a reference based on response variability.

It is useful when comparing performance across datasets or contexts with different scales.

The chapter notes that $\text{RelMSE} = 0$ represents a perfect model under this measure.

A RelMSE greater than 1 indicates that the created model is not good according to the chapter's interpretation.

The reference/baseline should be defined consistently when computing the measure.

Worked Problem 2 – Relative MSE

Note: the slide makes the reference definition explicit because relative-error conventions can vary.

Use the validation example $y = [80, 75]$, $\hat{y} = [75, 85]$.

Mean of actual values: $\bar{y} = (80+75)/2 = 77.5$.

Using the population-variance reference for the constant-mean predictor:

Baseline MSE = $[(80-77.5)^2 + (75-77.5)^2]/2 = 6.25$.

Model MSE = 62.5.

RelMSE = $62.5/6.25 = 10$.

Interpretation: the model's squared error is 10 times the constant-mean baseline for this tiny example.

Coefficient of Variation (CV)

Textbook p. 140

CV compares variability with the mean.

Formula: $CV = \sigma/\mu$.

Percentage CV = $(\sigma/\mu) \times 100$.

It is useful when comparing variability for quantities measured on different scales.

CV has no physical unit.

A higher CV means greater variability relative to the mean.

Interpret CV together with the response distribution and application.

Worked Problem 3 – Calculate CV

Data: response values = [10, 12, 14, 16, 18].

Mean $\mu = (10+12+14+16+18)/5 = 14$.

Population variance = $[(-4)^2+(-2)^2+0^2+2^2+4^2]/5 = 8$.

Population standard deviation $\sigma = \sqrt{8} \approx 2.828$.

CV = $2.828/14 \approx 0.202$.

Percentage CV $\approx 20.2\%$.

Interpretation: standard deviation is about 20.2% of the mean.

Coefficient of Determination – r^2

Textbook pp. 140–141

r^2 measures the proportion of variation in the response accounted for by the regression relationship.

Formula: $r^2 = 1 - \text{SSE}/\text{SST}$.

$\text{SSE} = \sum (y_i - \hat{y}_i)^2$ = unexplained variation/error sum of squares.

$\text{SST} = \sum (y_i - \bar{y})^2$ = total variation.

Explained variation = $\text{SST} - \text{SSE}$.

Higher r^2 generally indicates stronger explanatory fit.

High r^2 does not prove causation and should be interpreted with error measures and model context.

Worked Problem 4 – Calculate r^2

Actual values: $y = [10, 12, 15, 18]$.

Predicted values: $\hat{y} = [11, 11, 14, 20]$.

Mean actual response: $\bar{y} = (10+12+15+18)/4 = 13.75$.

$SSE = (10-11)^2 + (12-11)^2 + (15-14)^2 + (18-20)^2 = 7$.

$SST = (10-13.75)^2 + (12-13.75)^2 + (15-13.75)^2 + (18-13.75)^2 = 37.00$.

$r^2 = 1 - 7/37 \approx 0.8108$.

About 81.1% of the total response variation is accounted for by the fitted relationship in this calculation.

Worked Problem 5 – Comparing Two Models

Suppose Model A has $MAE = 4.2$, $MSE = 30$ and $r^2 = 0.88$.

Model B has $MAE = 3.5$, $MSE = 24$ and $r^2 = 0.84$.

Which model is better?

For error metrics MAE and MSE, lower is better $\rightarrow$ Model B.

For r^2 , higher is better $\rightarrow$ Model A.

Conclusion: there is no single-number decision. Consider the purpose and balance predictive error with explanatory fit.

Problem 5 – Interpretation

Metric	Model A	Model B	Better by this metric
MAE	4.2	3.5	B
MSE	30	24	B
r^2	0.88	0.84	A
Overall	Depends on objective	Depends on objective	Use validation context

Standard Error Estimate

Textbook p. 141

The chapter discusses standard error estimation using squared differences between observed and predicted values.

It quantifies the residual spread around the fitted regression relationship.

It gives an indication of the variability left unexplained by the model.

A smaller residual spread generally indicates predictions closer to the fitted relationship.

Do not confuse residual standard error with standard errors associated with individual regression coefficients.

Worked Problem 6 – Residual Error Interpretation

Suppose a regression model produces residuals: $[2, -1, 3, -2, 0]$.

$$SSE = 2^2 + (-1)^2 + 3^2 + (-2)^2 + 0^2 = 18.$$

$$MSE = 18/5 = 3.6.$$

$$RMSE = \sqrt{3.6} \approx 1.897.$$

Interpretation: the prediction error has a typical RMSE scale of about 1.90 response units.

The residuals should also be examined for systematic patterns rather than relying only on one metric.

Worked Problem 7 – Complete Validation

Actual: $y = [20, 25, 30, 35, 40]$.

Predicted: $\hat{y} = [22, 24, 29, 37, 39]$.

Residuals = $[-2, +1, +1, -2, +1]$.

MAE = $(2+1+1+2+1)/5 = 1.4$.

SSE = $4+1+1+4+1 = 11$.

MSE = $11/5 = 2.2$.

RMSE = $\sqrt{2.2} \approx 1.483$.

Problem 7 – Calculate r^2

Actual mean: $\bar{y} = (20+25+30+35+40)/5 = 30$.

$SST = (20-30)^2 + (25-30)^2 + (30-30)^2 + (35-30)^2 + (40-30)^2$.

$SST = 100 + 25 + 0 + 25 + 100 = 250$.

$SSE = 11$.

$r^2 = 1 - 11/250 = 0.956$.

Interpretation: approximately 95.6% of the response variation is accounted for by the fitted relationship.

Validation Metrics – Formula Sheet

Metric	Formula / definition
Residual	$e_i = y_i - \hat{y}_i$
MAE	$(1/n) \sum y_i - \hat{y}_i $
MSE	$(1/n) \sum (y_i - \hat{y}_i)^2$
RMSE	$\sqrt{\text{MSE}}$
RelMSE	Prediction error relative to a specified reference/baseline
CV	σ/μ ; percentage CV = $(\sigma/\mu) \times 100$
SSE	$\sum (y_i - \hat{y}_i)^2$
SST	$\sum (y_i - \bar{y})^2$
r^2	$1 - \text{SSE}/\text{SST}$

MAE vs MSE vs RMSE

MAE: easiest to interpret because it is in the original response units.

MSE: strongly penalizes large errors because residuals are squared.

RMSE: square root of MSE, returning to original response units.

If large errors are especially undesirable, MSE/RMSE are informative.

If equal absolute error treatment is preferred, MAE is useful.

Use more than one metric when evaluating a practical regression model.

How to Choose a Validation Metric

Need a direct average error in original units → MAE.

Need to penalize large errors strongly → MSE.

Need squared-error information but in original units → RMSE.

Need comparison relative to a baseline → Relative MSE.

Need variability relative to mean → CV.

Need proportion of variation explained → r^2 .

Need residual spread around the fitted relationship → standard error estimate.

For a complete assessment, report several complementary measures.

Exam Problem 8 – L3 Numerical

Actual values: [50, 60, 70, 80].

Predicted values: [52, 58, 72, 77].

Find: (a) residuals, (b) MAE, (c) MSE, (d) RMSE.

Residuals: [-2, +2, -2, +3].

$MAE = (2+2+2+3)/4 = 2.25$.

$SSE = 4+4+4+9 = 21$.

$MSE = 21/4 = 5.25$.

$RMSE = \sqrt{5.25} \approx 2.291$.

Exam Problem 9 – L4 Model Evaluation

Model A: MAE=5, RMSE=8, $r^2=0.91$.

Model B: MAE=4, RMSE=6, $r^2=0.87$.

Question: Which model should be selected for a forecasting application where prediction error is the main concern?

Solution: Model B has lower MAE and RMSE, so it provides smaller prediction errors.

Although Model A has higher r^2 , Model B may be preferred when accurate predictions are the primary objective.

Final selection should also consider validation/test data and application requirements.

Common Student Mistakes in 5.4

Using signed residuals directly for MAE.

Forgetting to square residuals for MSE.

Reporting MSE as if it were in the original units.

Confusing MSE and RMSE.

Interpreting high r^2 as proof that a model is causal.

Comparing CV values without considering the means and distributions.

Using r^2 alone without examining prediction error.

Confusing residual standard error with coefficient standard error.

Using a relative metric without clearly identifying its reference/baseline.

Integrated Problem 10 – Full Validation Workflow

A model predicts [105, 118, 131, 145] for actual values [100, 120, 130, 150].

Step 1: Calculate residuals = [-5,+2,-1,+5].

Step 2: MAE = (5+2+1+5)/4 = 3.25.

Step 3: SSE = 25+4+1+25 = 55.

Step 4: MSE = 55/4 = 13.75.

Step 5: RMSE = $\sqrt{13.75} \approx 3.708$.

Step 6: Calculate $\bar{y} = 125$; SST = 250+25+25+625 = 925.

Step 7: $r^2 = 1 - 55/925 \approx 0.9405$.

Interpretation of Problem 10

MAE = 3.25: average absolute prediction error is 3.25 response units.

RMSE $\approx$ 3.71: the squared-error-based error scale is about 3.71 response units.

$r^2 \approx 0.9405$: about 94.05% of response variation is accounted for by the fitted relationship.

The metrics suggest a relatively strong fit for this small dataset.

However, model quality should be established using appropriate validation/test data and domain context.

5.4 – Key Takeaways

Validation determines how well predicted values agree with actual values.

MAE measures average absolute error.

MSE measures average squared error and emphasizes large errors.

RMSE is the square root of MSE and is expressed in response units.

Relative MSE compares model error with a defined reference.

CV measures variability relative to the mean.

r^2 measures the proportion of response variation explained by the regression relationship.

Standard error estimate describes residual spread.

No single metric should be used blindly; select and interpret metrics according to the application.

Important Examination Questions

L2: Define MAE, MSE and RMSE and explain their differences.

L2: Explain coefficient of variation and coefficient of determination.

L3: Calculate MAE, MSE and RMSE for a given set of actual and predicted values.

L3: Calculate SSE, SST and r^2 for a regression model.

L3: Calculate CV for a given response dataset.

L4: Compare two regression models using MAE, RMSE and r^2 .

L4: Explain why a model with higher r^2 may not always be the preferred predictive model.

L4: Design a validation strategy for a real-world regression application.

Quick Revision – 5.4

Actual → Predicted → Residuals → Validation metrics → Interpretation.

MAE → absolute errors.

MSE → squared errors.

RMSE → square root of MSE.

RelMSE → relative error against a reference.

CV → standard deviation relative to mean.

r^2 → explained variation / total variation.

Standard error estimate → residual spread.

Best practice: combine error metrics, goodness-of-fit measures and application context.